

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. Examination (IT)

May 2022

IT380 Cyber Security

Date: 14.05.2022, Saturday

Time: 10:00 AM To 01:00 PM

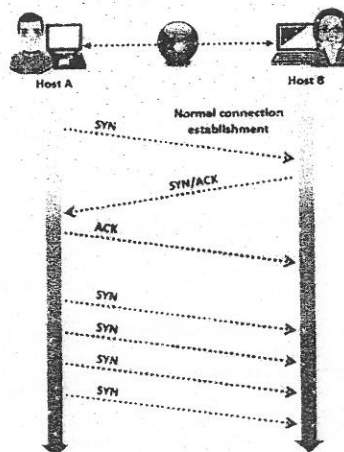
Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures/chart wherever required.

SECTION – I

- Q - 1** Define any seven terms: [07]
- i) Salami Attack ii) Phishing iii) Vishing iv) Web Jacking
v) Cyber Bullying vi) Cyber Stalking vii) Cyber Terrorism viii) Slack Space
- Q - 2** Attempt any two:
- (a) What is OWASP? Explain any three vulnerabilities from OWASP Top 10 vulnerabilities with example. [07]
- (b) Explain NIST framework for Cybersecurity. [07]
- (c) i) What is malware? What are the different ways for malware to enter the system? [03]
ii) Explain port scanning using Nmap. Explain below commands of Nmap. [04]
nmap -iL nmaptest.txt
- Q - 3 (a)** Define hacking and differentiate white hats, black hats and gray hats hackers. [07]
- (b) Elaborate stages of virus life cycle. [07]
- OR**
- Q - 3 (a)** Explain hacking phases. [07]
- (b) Explain below DOS attack. [07]



SECTION – II**Q – 4** Answer the following.

- i) In a case of manipulation, employees of one of the well-known telecom service provider were taken into custody in relation to the tampering of the electronic 32-bit number (ESN) that is programmed into cell phones. The theft was for another well-known telecom service provider. In a verdict on a later date, the court punished accused due to the source code manipulated. Recreate situation and answer that what punishment is applicable under which IT Act? [02]
- ii) In the case of fraud identity and unauthorized access, one management trainee of the bank was engaged to be married. The couple exchanged many emails using the company computers. After some time, the two broke up and the girl created fraudulent email ids and sent emails to the boy's foreign clients. She used the bank's computer to do this. The boy's company lost a large number of clients and took the bank to court. The bank was held liable for the emails sent using the bank's system. Justify the situation and answer that what punishment is applicable under which IT Act? [02]
- iii) Staff should not be told what monitoring is happening except in exceptional circumstances. Is it true or false? [01]
- iv) Directions under sub-section (6) of section 70B of the IT Act, 2000, the Indian government recently passed a rule requiring all the VPN service providers to collect and store user data for up to _____ years. [01]
- v) Differentiate cyber terrorist and hacktivist. [01]

Q - 5 Attempt any two:

- (a) Explain different types of firewalls. [07]
- (b) How network forensics can be done using Wireshark? Explain in detail. Explain this filter in wireshark. i) eth.dst == ff:ff:ff:ff:ff:ff ii) eth.addr == 00:70:f4:23:18:4c [07]
- (c) i) Explain these laws and sections in your words: section 65-tampering with computer source document, and section 67- publishing information which is obscene in electronic form. [05]
- ii) What is VPN and Tor? [02]

- Q – 6 (a)** Explain pre-investigation, investigation and post-investigation phases of investigation process. [07]
- (b) Which are rules admissible in the court of law as digital evidence? [07]

OR

- Q – 6 (a)** What is forensic readiness? Elaborate forensic readiness planning. [07]
- (b) Define anti-forensics in the domain of digital forensics and discuss any three anti-forensics techniques. [07]
